

Molecular Modeling and Simulations

Mini Project: Escaping the Pentane Barrier

1. Project Overview

Molecular simulations often become trapped in local free-energy minima and therefore fail to explore the full conformational space of a system within practical simulation times. This project investigates this issue using **n-pentane** as a model system.

The central backbone of n-pentane exhibits torsional flexibility, and rotation about the backbone dihedral gives rise to multiple conformational states separated by energy barriers. In a straightforward simulation, the system may remain confined to a single well for a long time, thereby providing a clear example of poor sampling.

The aim of this project is to develop a Python-based workflow that first demonstrates this trapping behavior and then introduces an enhanced sampling strategy that improves barrier crossing and conformational exploration.

Students must use a **united-atom (UA)** representation of n-pentane. In a UA model, hydrogen atoms attached to carbon are not treated as separate interaction sites; instead, they are incorporated into the carbon centers. This reduces the total number of particles and lowers the computational cost while preserving the essential torsional behavior of the molecule. The model must be parameterized using the published **TraPPE-UA** force field. Students should refer to the original paper at this link and to the parameter database available at <http://trappe.oit.umn.edu/UA>.

The workflow must include both **Monte Carlo (MC)** sampling and **NVT molecular dynamics (MD)**.

2. Problem Statement

For an **n-pentane** molecule represented using the **TraPPE-UA** force field, a naive simulation may remain trapped in one conformational region because of torsional energy barriers.

Your task is to:

- generate an initial configuration for the molecule,
- demonstrate baseline conformational sampling using MC and NVT MD,
- compare the dynamics at **120 K** and **250 K**,
- compare the **dihedral distributions** obtained from MC, MD, and enhanced sampling,
- compute the **potential of mean force (PMF)** using the **Boltzmann inversion** formula,
- implement an enhanced sampling method that improves barrier crossing,
- quantify the improvement in conformational exploration.

All implementations must be written in **Python**.

The use of standard molecular dynamics engines is not permitted. This includes, but is not limited to:

- GROMACS
- LAMMPS
- AMBER

- NAMD
- OpenMM

Standard scientific Python libraries such as NumPy, SciPy, and Matplotlib may be used.

3. Required Conditions for All Submissions

All submissions must satisfy the following conditions:

- Molecular system: **n-pentane**
- Representation: **united-atom (UA)**
- Force field: **TraPPE-UA**
- Reference paper: TraPPE-UA original paper
- Parameter source: TraPPE-UA parameter database
- Dihedral of interest: backbone dihedral defined by atoms **1–2–3–4**
- Temperature: **120 K and 250 K**
- MD ensemble: **NVT**
- Simulation length: **200,000 steps**
- Dihedral discretization: **36 equal bins** over -180° to 180°

Students must show that the baseline simulation exhibits trapping before applying the enhanced sampling method.

4. Task 1: Initial Configuration Generation (10 Marks)

Students must generate a physically reasonable initial configuration of n-pentane in the united-atom representation.

The initial configuration should be prepared in Python and should serve as the starting point for both the Monte Carlo and NVT MD simulations. The configuration must be consistent with the chosen TraPPE-UA model and the dihedral definition specified above.

5. Task 2: Baseline Sampling Using MC and NVT MD (25 Marks)

Students must implement baseline **Monte Carlo** and **NVT MD** simulations in Python using the TraPPE-UA force field.

The purpose of this stage is to demonstrate the limitations of naive sampling. The simulations should be performed at both **120 K** and **250 K**. Students should compare the dynamics at the two temperatures and discuss how temperature affects barrier crossing, conformational transitions, and trapping in local minima. The submitted analysis should include the dihedral angle time series, a histogram of sampled states, comparison of the dihedral distributions obtained from MC and MD, evidence that barrier crossing is infrequent, and a comparison between the MC and NVT MD behavior at the two temperatures.

6. Task 3: Enhanced Sampling Method (40 Marks)

Students must design and implement an enhanced sampling algorithm that improves exploration of the torsional landscape and enables barrier crossing.

The method must be implemented entirely in Python and integrated into the simulation workflow. The report should clearly explain the algorithmic design, how it is incorporated into the simulation, and why it is expected to improve sampling relative to the baseline methods.

The submitted results should demonstrate improved conformational exploration. Students should also compare the dihedral distributions obtained from the enhanced method against those from the baseline MC and MD simulations.

7. Task 4: Sampling Efficiency and Free-Energy Analysis

(25 Marks)

The performance of the method will be evaluated using entropy-based measures that quantify conformational exploration.

Exploration Entropy

Let the dihedral angle be divided into 36 bins. If P_i denotes the probability of observing bin i , the exploration entropy is defined as

$$S = - \sum_i P_i \log(P_i)$$

Higher entropy indicates broader exploration of conformational space.

Early Exploration Score

Let $S(t)$ denote the entropy computed using samples collected up to step t . The early exploration score is defined as

$$E = \frac{1}{T} \sum_{t=1}^T S(t)$$

where T is the total number of simulation steps.

Higher values of E indicate that new conformations are discovered earlier during the simulation.

Potential of Mean Force

Using the sampled dihedral distribution, students should compute the **PMF** by applying the Boltzmann inversion formula,

$$F(\phi) = -k_B T \ln P(\phi) + C$$

where $P(\phi)$ is the normalized dihedral probability distribution, k_B is the Boltzmann constant, T is the temperature, and C is an arbitrary constant.

The PMF should be reported for the baseline and enhanced sampling methods, and the free-energy profile should be discussed in relation to the observed conformational states.

8. Submission Requirements

Each submission must include:

- Python code for initial configuration generation,
- Python code for the baseline MC and NVT MD simulations,
- Python code for the enhanced sampling method,
- plots and analysis of the simulation results,
- a written report describing the methodology, implementation, and conclusions.